

Faculty Research Grant & Publication Tracker — Layered Architecture

UML Component Diagram · J Karthik (PES1UG25AM804) · Lab 3

PRESENTATION LAYER

Web Portal UI

Faculty Researcher screens · Research Dean dashboard

Browser client

BUSINESS LAYER

Grant & Expense Manager

Grant records · expense recording
expense-cap validation · remaining balance

IGrantService

Publication Manager

Publication records · indexing · citations
publication metrics

IPublicationService

Approval Workflow

Co-author approval requests
approval review · status tracking

IApprovalReview

IApprovalRequest

Analytics Service

Fund burn-up analytics
research & publication performance metrics

AnalyticsService

DATA LAYER

Database Repository

grants · expenses · publications
users · parameterised data access

IDataAccess

IAuditLog

Audit Ledger

Immutable, tamper-evident audit records
append-only logging

Legend:

○ Provided interface

⊂ Required interface

— Assembly / dependency

- - - Dashed = audit event